

Lachlan Astfalck

Curriculum Vitae

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Education

2016–2019 **Doctor of Philosophy**, *Faculty of Engineering, Mathematics and Computing, The University of Western Australia.*

Conferred June 2019

2010–2014 **Bachelor of Mechanical Engineering**, *The University of Western Australia.*

First Class Honours, **Valedictorian**

Doctoral Thesis

Title *Quantification and propagation of meteorological and oceanographic uncertainties in offshore engineering*

Supervisors Dr. Edward Cripps, Dr. Ian Milne & Prof. Melinda Hodkiewicz

Description This thesis quantified long-term uncertainties related to wind, surface current, and waves, and propagated the uncertainties through computationally demanding simulators of vessel motions with application to vessels moored off Western Australia. This thesis was graded in the top 5% of PhD theses at UWA.

Research Employment

2022–Present **Research Fellow**, *THE UNIVERSITY OF WESTERN AUSTRALIA*, Perth, Australia.

I am a research statistician within the ARC ITRH for Transforming energy Infrastructure through Digital Engineering (TIDE), which provides mathematical and data-driven solutions to modern engineering problems. Methodologically, I am focused on developing non-parametric statistical methodology for spectral, temporal and spatio-temporal analysis; some of this work has been published on two occasions in *Biometrika*. I work across many applied disciplines, most notably in fine-scale oceanography and the hydrodynamics of moored assets. Within this capacity I work frequently with multiple and varied shareholders; including, applied scientists, local industry, and international mathematical collaborators.

2020–2022 **Research Fellow**, *THE UNIVERSITY OF LEEDS*, Leeds, United Kingdom.

This work, in conjunction with climate scientists, developed rigorous methods of uncertainty quantification for large non-standard spatio-temporal computer models with application to past deglaciation events. A number of novel statistical methods were developed as part of the project, including fast and efficient methodology for reconstructions of spatio-temporal fields, emulation of non-standard computer models, and some extensions to Bayes linear statistics. Some of these results are published in the *Journal of the American Statistical Association*. As a result of my input to this project, I remained a co-investigator for the project duration.

2019 **Research Associate**, *THE UNIVERSITY OF WESTERN AUSTRALIA*, Perth, Australia.

This work was a consulting position that made much of the research conducted in my PhD thesis operational. This included temporal modelling of directional wave spectra, temporal modelling of wind, and spatio-temporal modelling of surface current. These uncertainty quantifications were used to drive emulators of vessel motion models to provide near-real time probabilistic predictions. The majority of this work was completed with local industry.

Teaching Appointments

- STAT2063 **Probabilistic Methods and their Applications**, *Unit Coordinator*, UWA.
Led coordination of a core second-year statistics unit for engineering students, including lecture delivery, curriculum development, assessment design, sessional staff management, and student support.
- DATA5710 **Applied Statistics for Complex Data**, *Lecturer*, The University of Sydney.
Delivered intensive postgraduate coursework in applied statistics as part of the ARC Training Centre in Data Analytics for Resources and Environments.
- GENG5507 **Risk, Reliability and Safety**, *Lecturer & Tutor*, UWA.
Taught and co-developed material for a postgraduate engineering unit covering probabilistic risk analysis and statistical reliability methods; responsibilities included lecturing, tutoring, and assessment.
- MATH2032 **Complex and Fourier Analysis**, *Guest Lecturer*, UWA.
Guest lecture and workshop on the practical applications of Fourier methods.
- ENSC3012 **Data Collection and Analysis**, *Guest Lecturer*, UWA.
Guest lecture on introductory statistical and machine learning methods.
- GENG1002 **Introduction to Electrical Engineering (now ENSC2003)**, *Tutor*, UWA.
Provided tutorials and marked assessments in a first-year core engineering unit, with a focus on mathematical foundations and circuit analysis.
- GENG1001 **Introduction to Engineering Mechanics (now GENG1010)**, *Tutor*, UWA.
Provided tutorials and marked assessments in foundational engineering mechanics.

Refereed Publications

- 2025 **LC Astfalck**, EJ Cripps, and AM Sykulski. Bias correction of quadratic spectral estimators. *Biometrika*, asaf033.
- 2025 MD Rayson, **LC Astfalck**, ALS Ponte, AP Zulberti, and NL Jones. Characteristic velocity and timescales of non-phase-locked internal tides in a mesoscale eddy field. *Journal of Geophysical Research: Oceans*, 130(6), e2024JC021789
- 2025 IA Milne, **LC Astfalck**, M Zed, and J Lee-Kopij. Response-based forecasting of vessel motion: From physics-based to data-driven methods. *Offshore Technology Conference*, D011S015R002
- 2024 **LC Astfalck**, EJ Cripps, and AM Sykulski. Debiasing Welch's method for spectral density estimation. *Biometrika*, 111(4), 1313–1329.
- 2024 **LC Astfalck**, DB Williamson, N Gandy, LJ Gregoire, and RF Ivanovic. Coexchangeable process modelling for uncertainty quantification in joint climate reconstruction. *Journal of the American Statistical Association*, 119(547), 1751–1764..
- 2024 ALS Ponte, **LC Astfalck**, MD Rayson, AP Zulberti, and NL Jones. Inferring flow energy, space and time scales: freely-drifting vs fixed point observations. *Accepted in Nonlinear Processes in Geophysics Discussions*, 31(4) 571–586.
- 2024 RY Zhang, HB Moss, **LC Astfalck**, EJ Cripps, and DS Leslie. BALLAST: Bayesian active learning with look-ahead amendment for sea-drifter trajectories. *NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty*.
- 2024 IA Milne, and **LC Astfalck**. Heave response spectra for a semisubmersible in long period swell. *24th Australasian Fluid Mechanics Conference*.
- 2024 N Gandy, **LC Astfalck**, GL Ives, and GE Rivers. Ice sheet speed-dating: using expert judgement to identify “good” simulations of the LGM North American ice sheets. *Quaternary Science Reviews*, 333:108690.

- 2024 VL Patterson, LJ Gregoire, RF Ivanovic, N Gandy, J Owen, RS Smith, PJ Valdes, OG Pollard and **LC Astfalck**. Contrasting the Penultimate and Last Glacial Maxima (21 and 140 ka BP) using coupled climate-ice sheet modelling. *Climate of the Past*, 20(10), 2191–2218.
- 2023 **LC Astfalck**, M Bertolacci, and EJ Cripps. Evaluating probabilistic forecasts for maritime engineering operations. *Data-Centric Engineering*, 4:e15.
- 2023 OG Pollard, NL Barlow, LJ Gregoire, N Gomex, V Cartelle, JC Ely, and **LC Astfalck**. Quantifying the uncertainty in the Eurasian ice-sheet geometry at the Penultimate Glacial Maximum (Marine Isotope Stage 6). *The Cryosphere Discussions*, 17(11):4751–4777.
- 2023 N Gandy, **LC Astfalck**, LJ Gregoire, RF Ivanovic, and VL Patterson. De-tuning a coupled climate ice sheet model to simulate the North American ice sheet at the Last Glacial Maximum. *Journal of Geophysical Research: Earth Surface*, 128(8):e2023JF007250.
- 2022 MA Loake, **LC Astfalck**, and EJ Cripps. Modelling sea surface wind measurements on Australia’s North-West Shelf. *Ocean Engineering*, 244:110308.
- 2020 M Masarei, **LC Astfalck**, AL Guzzomi, DJ Merritt, and TE Erickson. Soil rock content influences the maximum seedling emergence depth of a dominant arid zone grass. *Plant and Soil*, 450(1):497–509.
- 2019 **LC Astfalck**, EJ Cripps, MR Hodkiewicz, and IA Milne. A Bayesian approach to the quantification of extremal responses in simulated dynamic structures. *Ocean Engineering*, 182:594–607.
- 2019 **LC Astfalck**, EJ Cripps, JP Gosling, and IA Milne. Emulation of vessel motion simulators for computationally efficient uncertainty quantification. *Ocean Engineering*, 172:726–736.
- 2018 **LC Astfalck**, EJ Cripps, JP Gosling, MR Hodkiewicz, and IA Milne. Expert elicitation of directional metocean parameters. *Ocean Engineering*, 161:268–276.
- 2017 **LC Astfalck**, GK Kelly, X Li, and TB Sercombe. On the breakdown of SiC during the selective laser melting of aluminum matrix composites. *Advanced Engineering Materials*, 19(8):1600835.
- 2017 **LC Astfalck**, MR Hodkiewicz. Hamiltonian Monte Carlo sampling for Bayesian hierarchical regression in prognostics. *Asia Pacific Conference of the Prognostics and Health Management Society*.
- 2016 **LC Astfalck**, MR Hodkiewicz, A Keating, EJ Cripps, and M Pecht. A modelling ecosystem for prognostics. *Annual Conference of the Prognostics and Health Management Society*.
- 2016 J Sikorska, MR Hodkiewicz, A D’Cruz, **LC Astfalck**, and A Keating. A collaborative data library for testing prognostic models. *Prognostics and Health Management Society European Conference*.

Publications in Review

- 2025 **LC Astfalck**. Universal approximation of auto-covariance functions via spline kernels. *Submitted to Journal of the Royal Statistical Society: Series B (discussions)*.
- 2025 **LC Astfalck**, D Sen, S Patra, EJ Cripps and DB Dunson. Posterior projection for inference in constrained spaces. *Submitted to the SIAM/ASA Journal on Uncertainty Quantification*.

- 2025 T Tang, **LC Astfalck** and DB Dunson. Efficient Bayesian inference for continuous time Markov chains. *Submitted to the Journal of the American Statistical Association.*
- 2025 RY Zhang, HB Moss, **LC Astfalck**, EJ Cripps, and DS Leslie. Spatio-temporal active learning of time-dependent vector fields using Lagrangian observers. *Submitted to The Thirty-Ninth Annual Conference on Neural Information Processing Systems.*
- 2025 Y Zheng, MD Rayson, **LC Astfalck**, and NL Jones. A Machine Learning Parameterization for the Internal Gravity Wave Spectrum. *Submitted to the Journal of Geophysical Research: Oceans.*
- 2024 **LC Astfalck**, CR Bird, and DB Williamson. Generalised Bayes linear inference. *Submitted to Bayesian Analysis.*

Grants, Funding, Scholarships and Awards

- 2025 **ARC Discovery Project**, Quantifying critical ocean heat transport pathways at Ningaloo in a changing climate, Chief Investigator, ~ \$750 000AUD (EOI accepted, final decision October 2025)
- 2025 **UWA Shell Chair Funding**, Two years of research funding paid at Level B/C ~ \$250 000AUD
- 2024 **UWA Data Institute Summer Research Project Grant** Improved georeferencing of aerial imagery over the ocean, Co-Supervisor, \$8 000AUD
- 2024 **UWA Data Institute Summer Research Project Grant** Modelling Canopy Flow in Agrivoltaics Solar Panel Arrays using Computational Fluid Dynamics, Co-Supervisor, \$8 000AUD
- 2023 **TIDE Seed Funding** Bayesian time-series analysis of ocean drifter data, Principal Investigator, \$14 700AUD
- 2023 **TIDE Seed Funding** Remote sensing of surface currents from aircraft, Chief Investigator, \$15 000AUD
- 2021 **OceanWorks Prototyping Grant** Exploratory analysis of flying-lead failures, Principal Investigator, \$5 000AUD
- 2017–2019 **ARC OFFshore Hub postgraduate scholarship**, \$10 000AUD per annum top-up
- 2016–2019 **Research Training Program scholarship** (f.k.a. Australian Postgraduate Award), \$27 000AUD per annum
- 2014 **Valedictorian**, Mechanical Engineering, UWA
- 2014 **Chevron Chair Prize in Gas Process Engineering**
- 2012 **Convocation of UWA Graduate Association Undergraduate Prize**

Student and Research Associate Supervision

- 2023–Present William Edge, **Postdoctoral Research Associate**, advisor, UWA, In Progress
- 2024–Present Marcus Dyson, **PhD Candidate**, co-supervisor, UWA, In Progress
- 2024–Present Rui Zhang, **PhD Candidate**, co-supervisor, Lancaster University, In Progress
- 2022–Present Yutao Zheng, **PhD Candidate**, co-supervisor, UWA, In Progress
- 2024–Present Max Coten, **Honours Student**, principal supervisor, UWA, In Progress
- 2023–2024 Max Collins, **Graduate Research Associate**, principal supervisor, UWA, Completed.
- 2022–2023 Max Collins, **Honours Student**, co-supervisor, UWA, Completed
- 2022 Anthony D’Angelo, **Honours Student**, co-supervisor, UWA, Completed

- 2020 Muhammad Qasim, **Masters Student**, co-supervisor, University of Leeds, Completed
- 2020 Minjie Wu, **Masters Student**, co-supervisor, University of Leeds, Completed
- 2020 Max Anderson-Loake, **Honours Student**, co-supervisor, UWA, Completed (**subsequently awarded a Rhodes Scholarship**)

University Service and Industry Engagement

- 2025–Present Data-science Lead, Oceanworks Advisory Committee
- 2025–Present Member, Industry Working Group, Guidelines for Scour Assessment, Mitigation and Remediation of Subsea Structures
- 2024–Present Outreach Chair for UWA’s Department of Mathematics and Statistics
- 2024–Present Member, Engagement Advisory Group, ARC TIDE ITRH
- 2022–Present Events Chair for the ARC TIDE ITRH
- 2022–Present Ongoing industry engagement through the ARC TIDE ITRH
 - 2021 Development and presentation of three-part masterclass on Bayesian methods to local offshore industry
- 2016–2019 UWA UniMentor

Invited Presentations

- 2025 *Non-parametric ACF modelling for complex spatio-temporal processes*. Keynote, Environmental Section, Australian Statistical Conference.
- 2025 *The fast and the Fourier-ous: spectral bases for non-parametric modelling of the ACF*. Duke University.
- 2024 *Applications of Gaussian processes in oceanography*. Gaussian Process Summer School, The University of Manchester.
- 2024 *Some recent, and rediscovered, developments in Bayes linear statistics*. University of Exeter.
- 2023 *Debiasing Welch’s method of spectral density estimation*. Presented at Imperial College London, Lancaster University, and Durham University.
- 2022 *Probabilistic inference of nonlinear extreme ocean currents for operational forecasting*. UWA Data Science Forum.
- 2022 *Inferring nonlinear internal wave properties from sparse observations using statistical learning*. The Forum of Operational Oceanography.
- 2021 *Reconstructing boundary conditions for complex simulations with application to climate models*. Conference of the Royal Statistical Society.
- 2021 *Uncertainty, probability and decision making*. Bayesian Statistics and Forecasting for the Offshore Industry.
- 2021 *Hierarchical analysis and statistical analysis of computer codes*. Bayesian Statistics and Forecasting for the Offshore Industry.